

Input

Household-level attributes

- Household size, structure
- Vehicle ownership
- Income
- Housing type
- ...

Per-member attributes

- Age
- Gender
- Occupation
- License status
- Education level
- Relation to household head



1 Household-aware multi-task encoder

Integrate heterogeneous information and infer activity-pattern priors

Household-context encoder

Member attribute encoder (shared)

Set-attention experts (Progressive layered extraction)

Expert 1

Expert k

Expert K

Coordinated member representations

r_1

r_2

...

r_N

Joint input for decoder

Offline activity-pattern priors

Household-level activity-pattern prior

Individual-level activity-pattern prior

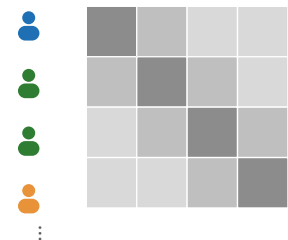
2 Household-coordinated activity-chain decoder

Generate activity chains of all members simultaneously

Start information for each valid member

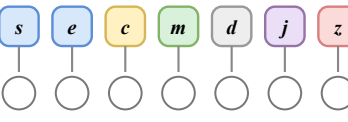
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Cross-role attention (Roles attend to co-members)



Transformer decoder layer ($\times L$)

Multi-attribute prediction heads (for each member at step t)



4 Nash-bargaining coordination layer

Enforce family-level feasibility and maximize household utility

Decoder outputs

□ □ □ □ ... □ □

Differentiable two-phase Nash bargaining optimization

Phase I: Feasibility projection

(Enforce family constraints)

- Joint travel & escorting
- Vehicle capacity & sharing
- Time & synchronization
- ...

Phase II: Utility maximization

(Maximize household utility)

- Individual utilities
- Fairness consideration
- Nash product maximization

Coordinated equilibrium solution

(Feasible & utility-maximized activity chains)

□ □ □ □ ... □ □

Output Household-level Activity Chains

Activity $a_{n,t} =$

- s departure time
- e arrival time
- c purpose
- m mode
- d driver flag
- j joint-travel flag
- z destination zone

Member 1



Member 2



Member 3



...

Member N



3 Spatially-aware destination attention layer

Select destination TAZ from a large candidate set

Activity context of member n at step t

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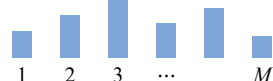
Spatially-aware destination attention

Geographic proximity

Purpose-specific attractiveness

Semantic similarity

Attention over candidate zones



Selected destination zone $z_{n,t}$

